

Computational Physics 2 Classes No.10 - Report

Diffusion equation: finite difference approach

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1 Introduction

The objective of this project was to solve the two-dimensional diffusion equation

$$\frac{\partial u}{\partial t} = \nabla \cdot (D(x, y) \nabla u) + S(x, y)$$

using a finite-difference discretization and explicit Euler time integration. The diffusion coefficient was assumed to vary spatially, making it necessary to employ a conservative flux formulation in order to correctly describe particle transport between neighboring cells.

The computational domain consisted of a square mesh covering the region

$$[-N, N] \times [-N, N],$$

with ($N=51$), resulting in a (103×103) grid. The spatial step was chosen as

$$\Delta x = \Delta y = 1,$$

while the time step was

$$\Delta t = \frac{1}{8}.$$

The diffusion coefficient was defined as

$$D = \begin{cases} 1, & \text{interior region,} \\ 0.005, & \text{intermediate layer of width 5 cells,} \\ 0.0001, & \text{outer layer of width 5 cells.} \end{cases}$$

A localized source term was introduced at the center of the domain,

$$S(x, y) = s, \delta(i, 0) \delta(j, 0),$$

with source strength

$$s = 0.01.$$

Initially the concentration was zero everywhere,

$$u(x, y, 0) = 0.$$

Zero-flux (Neumann) boundary conditions were imposed at the boundaries, ensuring that particles could not leave the computational domain. The conservative discretization was based on calculating diffusive fluxes through cell interfaces and updating the concentration according to the net incoming flux.

The main goals of the simulation were:

- to investigate the temporal evolution of the concentration field,
- to study the influence of spatially varying diffusion coefficients,

- to verify mass conservation by comparing the numerical integral

$$I(t) = \int \int u(x, y, t), dx, dy$$

with the analytical prediction

$$I(t) = st.$$

2 Results

Spatial distribution of the diffusion coefficient

Figure 1 presents the spatial distribution of the diffusion coefficient used in the simulation. The central part of the domain is characterized by a high diffusivity ($D=1$), while two low-diffusivity layers are placed near the boundaries.

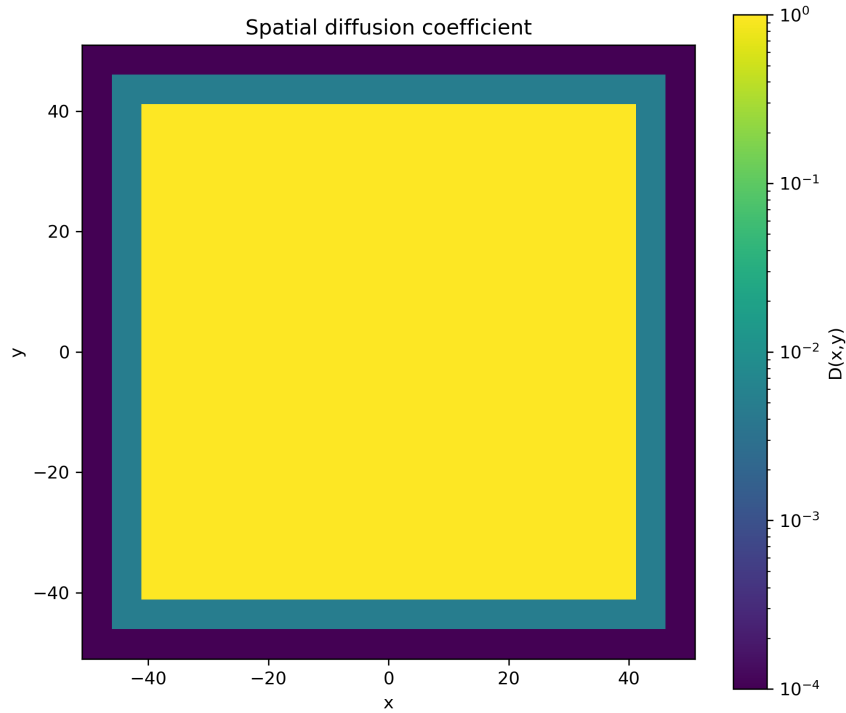


Figure 1: Spatial distribution of the diffusion coefficient.

The outermost layer has a diffusion coefficient four orders of magnitude smaller than the interior region. Consequently, particles reaching these regions diffuse very slowly. The intermediate layer ($D = 0.005$) acts as a transition region between the highly diffusive interior and the nearly insulating boundary.

Evolution of the concentration field

The temporal evolution of the concentration field is shown in Figure 2. Initially the concentration is equal to zero throughout the domain. As time progresses, particles are continuously generated by the source located at the center.

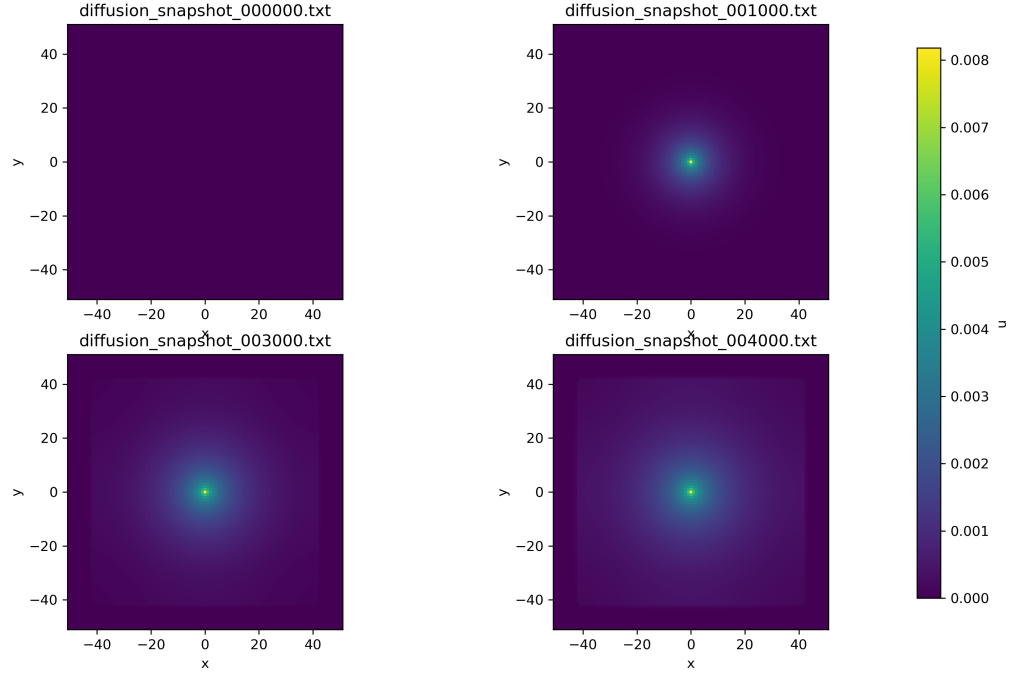


Figure 2: Evolution of the concentration field for selected simulation times.

At early times the concentration remains strongly localized around the source. Since the diffusion coefficient is equal to ($D=1$) in the interior region, particles spread rapidly away from the center and form an approximately radially symmetric distribution.

As the diffusion front approaches the low-diffusivity layers, the dynamics change significantly. Transport becomes increasingly suppressed and particles begin to accumulate near the interfaces separating regions with different diffusion coefficients. This effect is clearly visible in the later snapshots.

The concentration profile gradually loses its circular symmetry and develops a square-like shape. This behavior is a direct consequence of the geometry of the low-diffusivity regions. Because diffusion is strongly inhibited near the boundaries, the outer layers act as barriers that effectively confine particles inside the central part of the computational domain.

The final concentration profile therefore reflects not only the location of the source but also the spatial structure of the diffusion coefficient.

Mass conservation

An important verification of the numerical scheme is provided by the conservation of mass. Since the only source of particles is the central injection term and the boundaries are impermeable, the total amount of material inside the domain should satisfy

$$I(t) = st.$$

Figure 3 compares the numerical integral

$$I(t) = \int \int u(x, y, t), dx, dy$$

with the theoretical prediction.

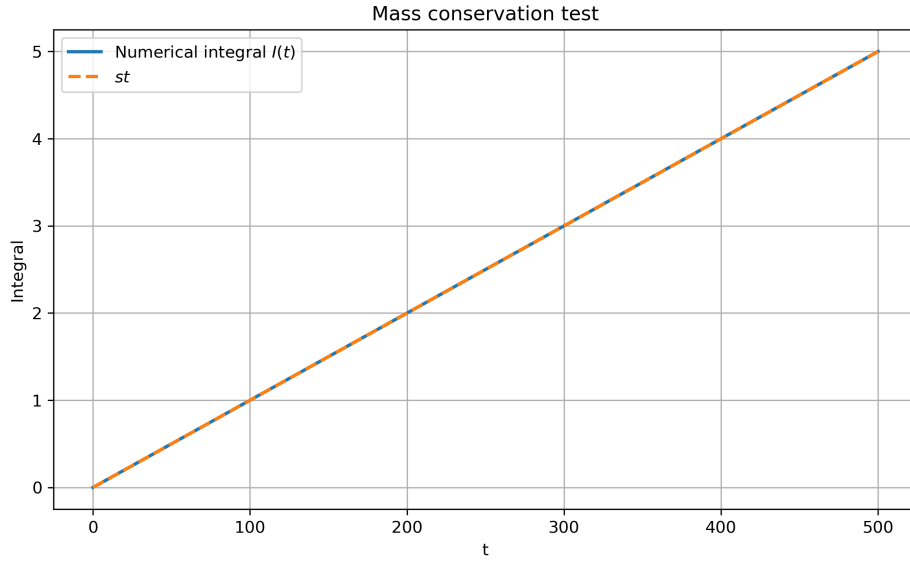


Figure 3: Comparison of the numerical integral with the analytical prediction (st).

The two curves overlap almost perfectly over the entire simulation interval. At the final simulation time ($t=500$), the theoretical value is

$$st = 0.01 \times 500 = 5,$$

which is reproduced by the numerical solution with extremely high accuracy.

To quantify the agreement, the relative error

$$\frac{I - st}{st}$$

was calculated and is shown in Figure 4.

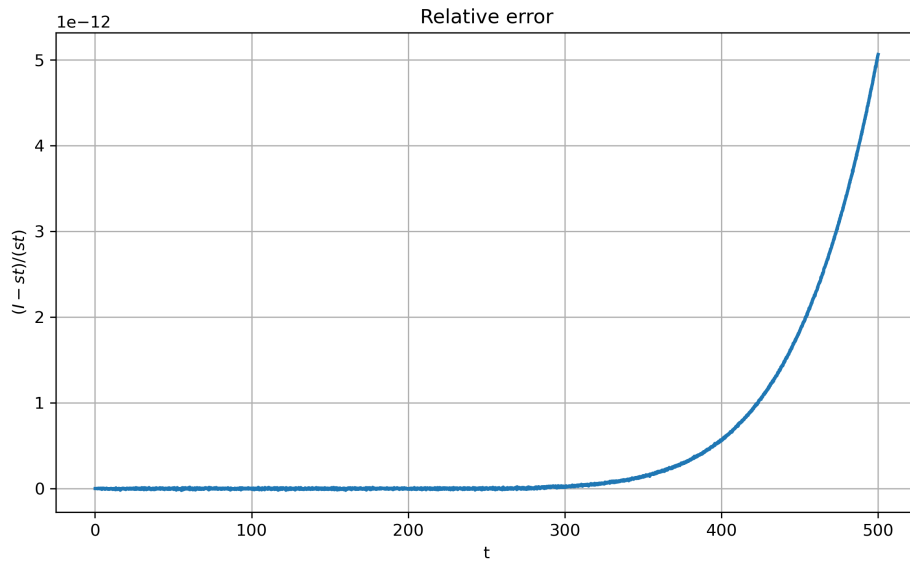


Figure 4: Relative error of the numerical integral.

The relative error remains below approximately

$$5 \times 10^{-12},$$

which is close to machine precision for double-precision floating-point arithmetic. Such a small error confirms that the conservative flux discretization correctly preserves the total mass of the system.

The excellent agreement between the numerical and analytical results demonstrates both the correctness of the implementation and the stability of the numerical method.

3 Conclusions

A finite-difference solution of the two-dimensional diffusion equation with spatially varying diffusion coefficient was successfully implemented. The use of a conservative flux formulation combined with explicit Euler time integration produced stable and physically meaningful results.

The simulations showed that diffusion is rapid in the central region where ($D = 1$), while transport becomes strongly suppressed in the low-diffusivity layers near the boundaries. As a result, the concentration profile evolves from an initially localized distribution into a broader structure whose shape is strongly influenced by the geometry of the diffusion coefficient field.

The numerical integral of the concentration was compared with the theoretical prediction ($I(t) = st$). The obtained agreement was excellent, with relative errors remaining below (5×10^{-12}). This confirms that the implemented finite-difference scheme preserves mass to machine precision.

Overall, the simulation successfully reproduced the expected physical behavior of diffusion in an inhomogeneous medium and verified the accuracy of the conservative finite-difference discretization.